

THORACIC WALL

Department of Anatomy

Thorax (1)

Second part

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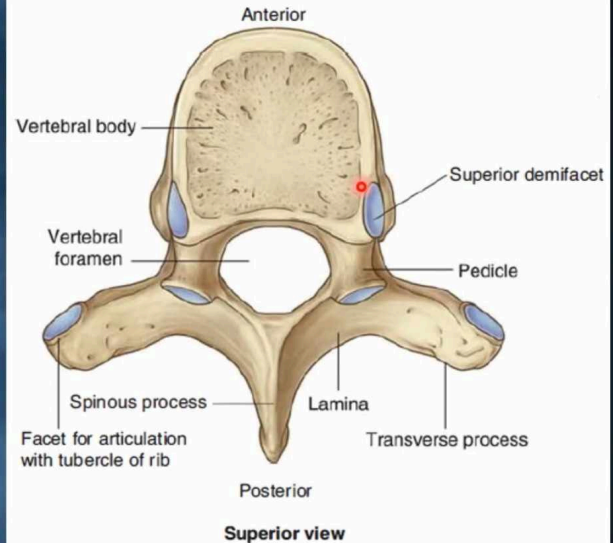
Lecture Points

- **Anatomy of the Thoracic wall and Thoracic Cavity.**
- **The superior thoracic aperture consists.**
- **The suprapleural membrane**
- **The bone Structure of the Thoracic Cage.**
 - **Sternum.**
 - **The sternal angle.**
 - **Ribs (Typical & Atypical).**
 - **Thoracic Vertebra (Typical & Atypical).**
 - **Joints (articulations) of the thorax**
- **Inferior Boundary (outlet) of the thorax.**
- **The diaphragm.**
 - ✓ **Physiological action of diaphragm.**
 - ✓ **Three Large Openings of Diaphragm.**
 - ✓ **arterial supply of the Diaphragm.**
- **The muscles of the thorax wall**
- **The arterial supply to the thoracic wall**
- **The venous drainage of the thoracic wall.**
- **lymphatic drainage of the thorax wall.**
- **Intercostal nerve (Typical & Atypical).**
- **Segmental innervation (dermatomes) of thoracic wall.**
- **herpes zoster the Spinal Ganglia**
- **Vertebral levels of sternum and transverse thoracic plane.**
- **Surface Anatomy of the Thoracic Wall Skeleton.**

Thoracic Vertebra

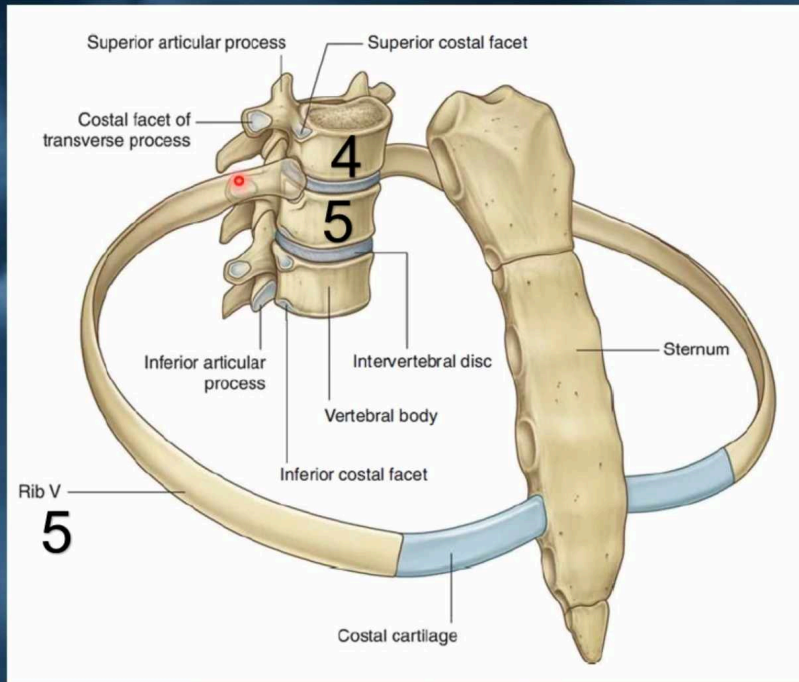
- **Heart-shaped** centrum (body)
- **Centrum notched** on **left** side for descending aorta
- **Intervertebral discs** = $\frac{1}{4}$ of total length of thoracic vertebral region
- **Superior and inferior costal demifacets:**

(on typical vertebrae (T2 – T9))

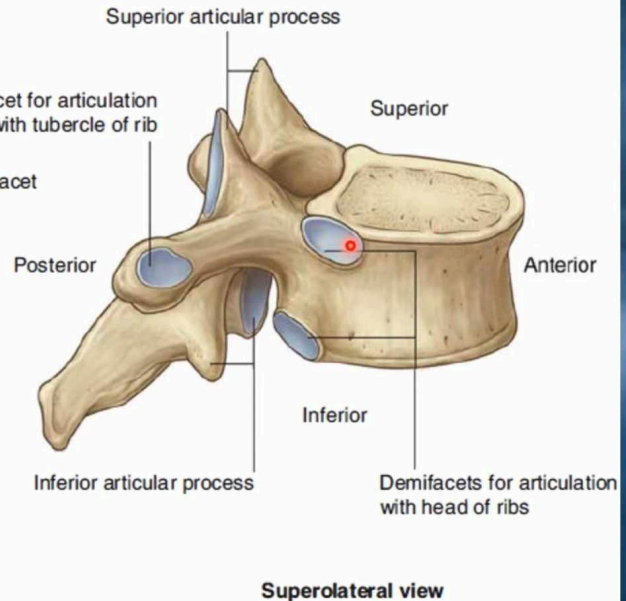
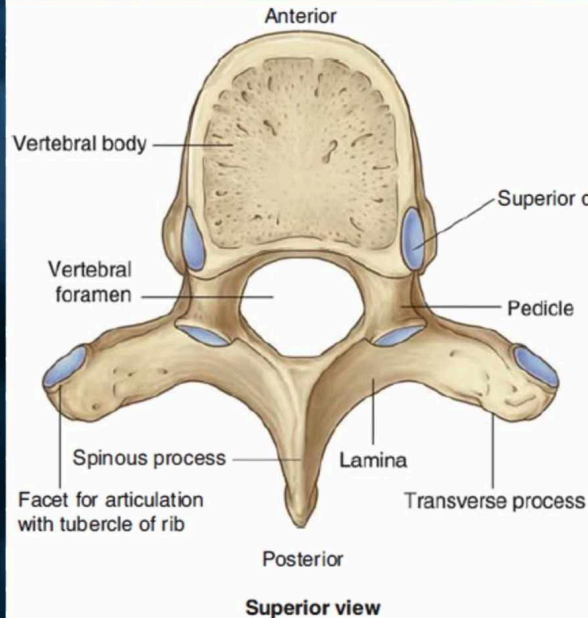


Typical rib articulation vertebrae

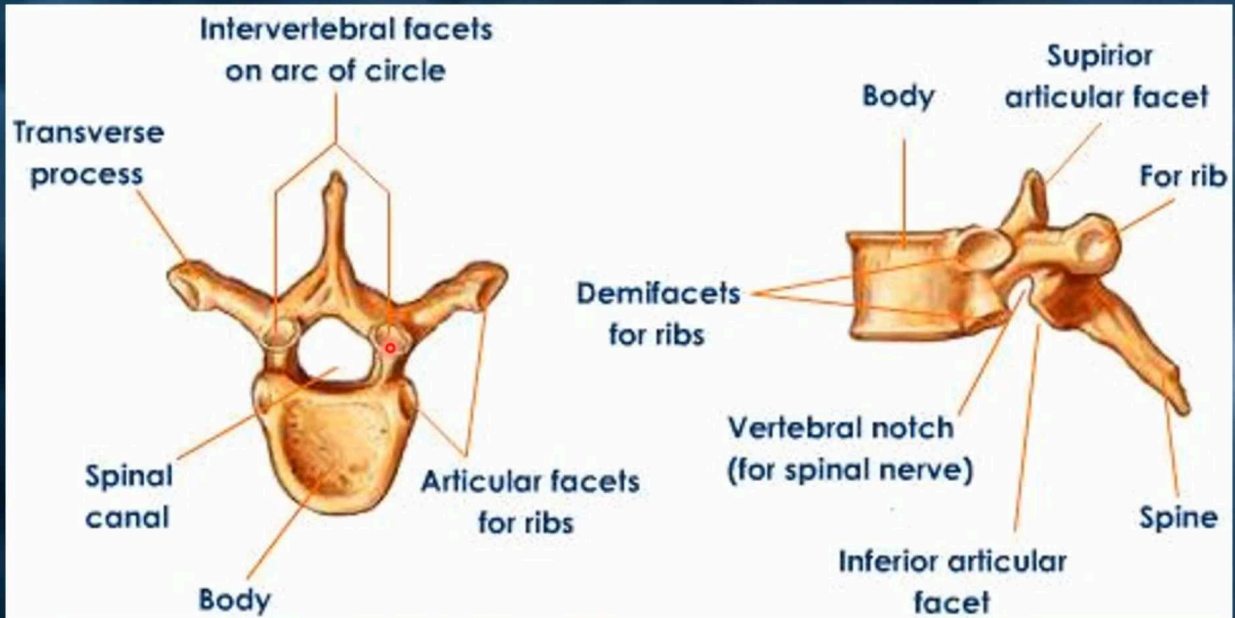
- **Head** articulates with **demifacets** of its own number **vertebra** and the **vertebra above** as well as the **intervertebral disc** between.
- **Tubercle** articulates with **transverse process** of same number vertebra.



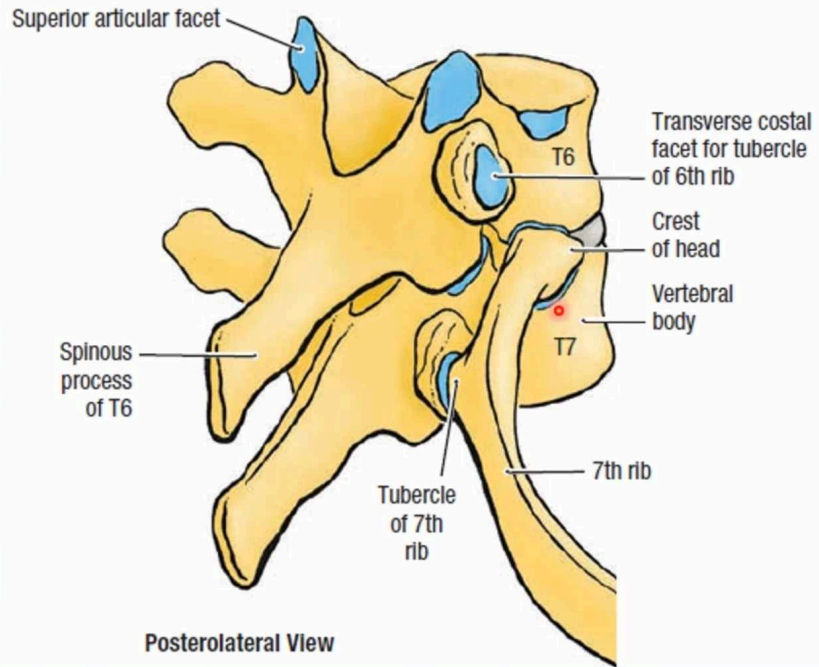
Typical Vertebrae



Typical Rib

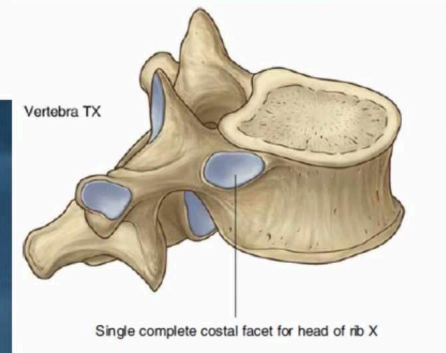
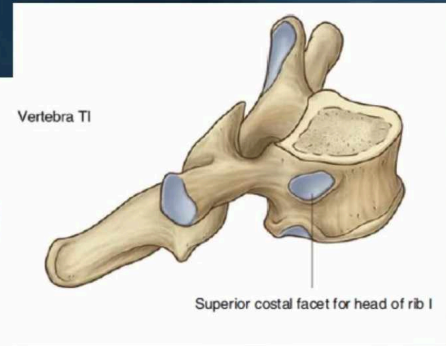
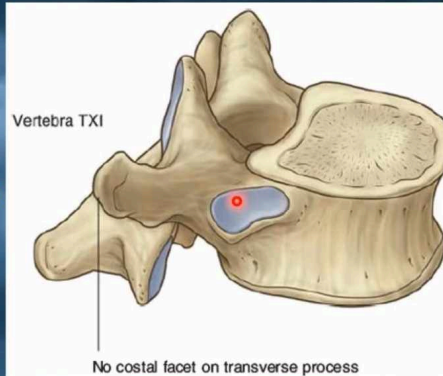


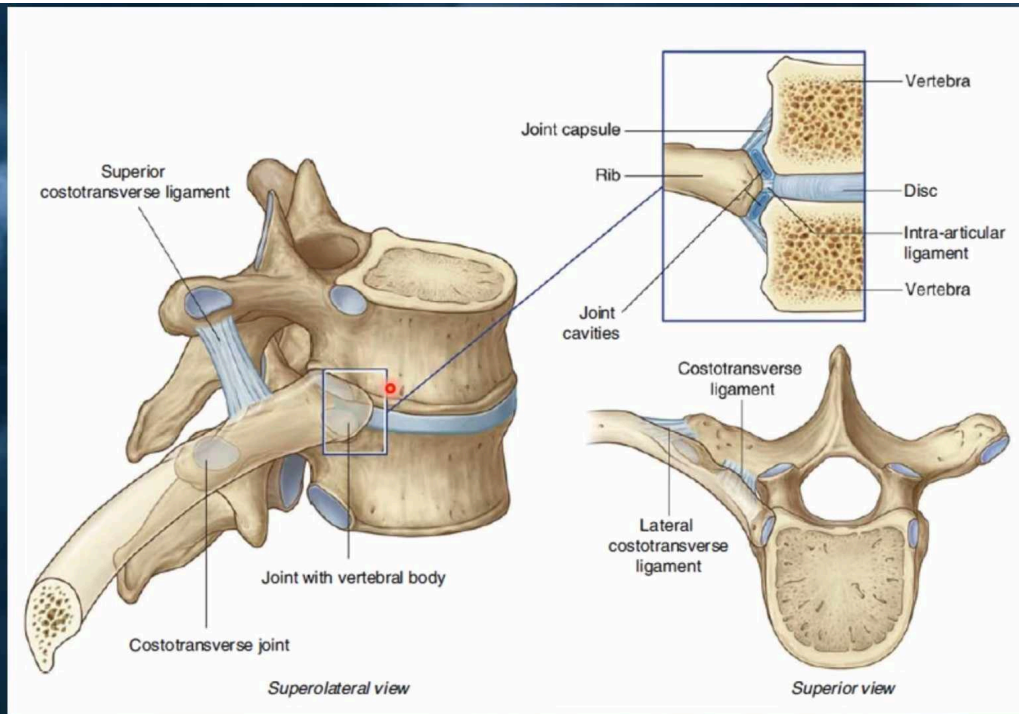
Typical Vertebrae



Atypical Thoracic Vertebra

- **T1:**
Superior costal facet
(not a demifacet).
- **T10:**
One pair of costal facets located **partly on body** and **partly on pedicle**.
- **T11 and T12:**
One pair of costal facets located on **pedicles**.





Articulations of the thorax

1) Sternoclavicular Joint:

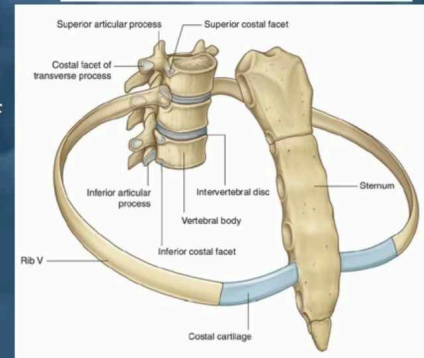
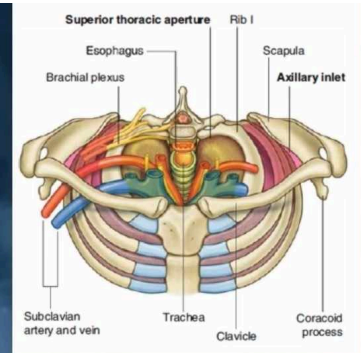
- **Sternal end of clavicle** with **manubrium of sternum** and **1st costal cartilage**.
- It is a **saddle-type synovial joint**.

2) Sternocostal (Sternochondral) Joints:

- first costal cartilages : **primary cartilaginous joint** (synchondrosis) Articulation of **first costal cartilages** with **manubrium of sternum**.
- 2nd-7th costal cartilages : **synovial plane joint** Articulation of **the 2nd-7th pairs of costal cartilages** with **sternum**.

3) Costochondral Joints

- It is **primary (hyaline) cartilaginous joint**.
- Are synchondroses in which **the ribs** articulate with their respective **costal cartilages**.



Articulations of the thorax

4) Costovertebral Joint:

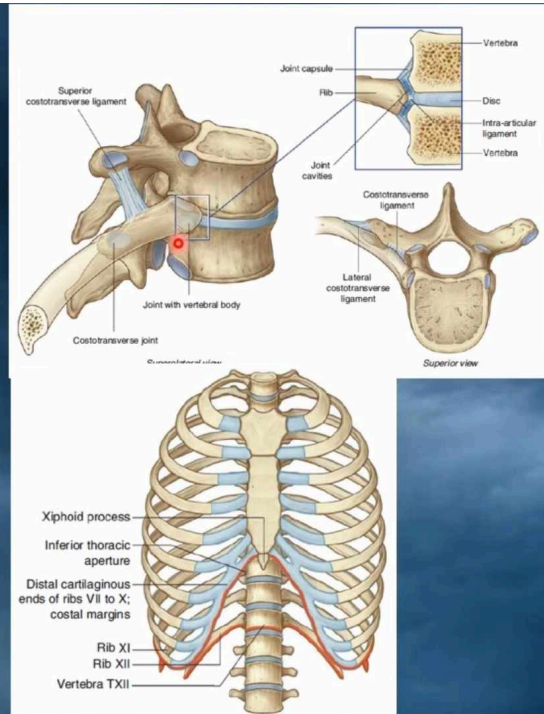
- Are **synovial plane joints** of **heads of ribs** with corresponding and **supraadjacent vertebral bodies**.

5) Costotransverse Joint:

- It is **synovial plane joint** of **tubercle of rib** with **transverse process of corresponding vertebra**.

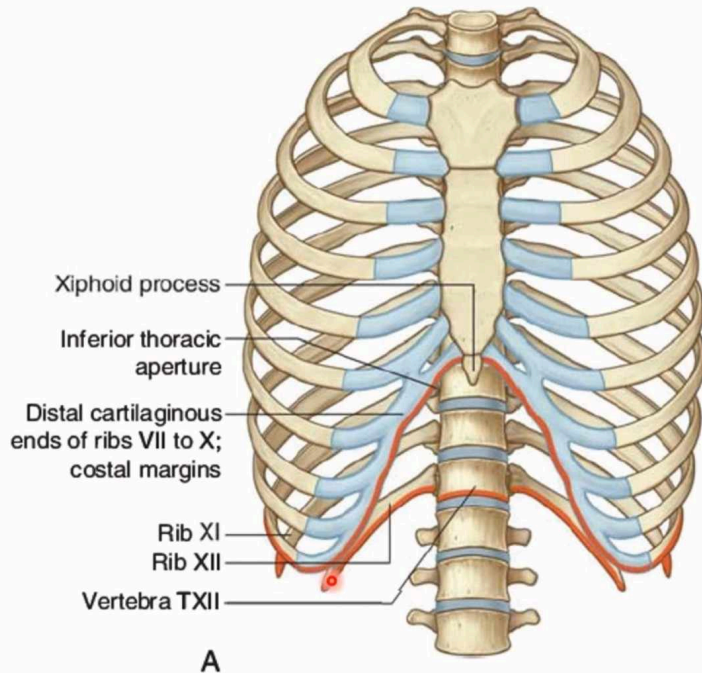
6) Interchondral Joint:

- Are **synovial plane joints** between **6th and 10th costal cartilages of ribs**.



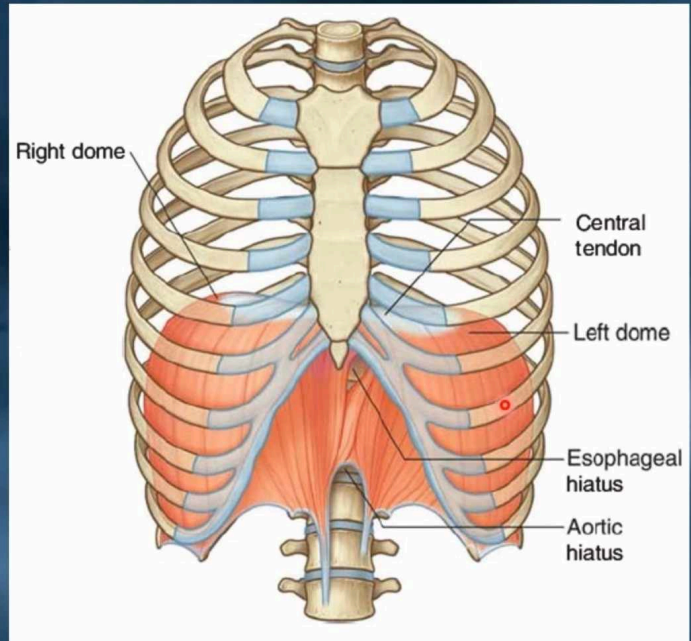
Inferior Boundary (outlet)

- **Twelfth thoracic vertebra.**
- Twelfth pair of ribs and costal cartilages.
- **Xiphisternal joint.**
- Closed by **the diaphragm.**
 - The diaphragm is pierced by a series of apertures to permit the passage of structures between the thorax and abdomen.



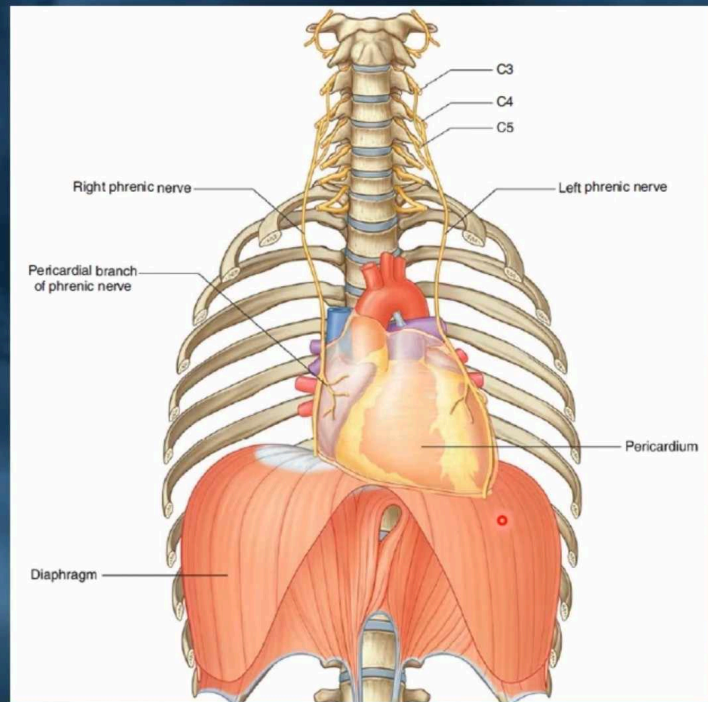
The diaphragm

- It is the **main muscle of respiration**.
- It is a **dome-shaped sheet of skeletal muscle** separating the thoracic and abdominal cavities and consisting of muscle fibres and a **central tendinous** portion.



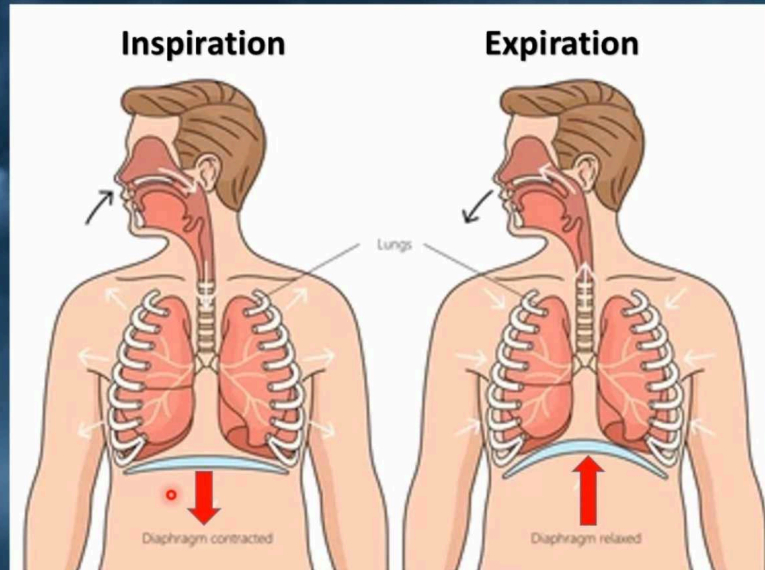
Innervation of the diaphragm

- It receives **somatic motor fibers** from the **phrenic nerve**.
- Its **central part** receives **sensory fibers** from the **phrenic nerve**.
- whereas the **peripheral part** receives **sensory fibers** from **the intercostal nerves**.



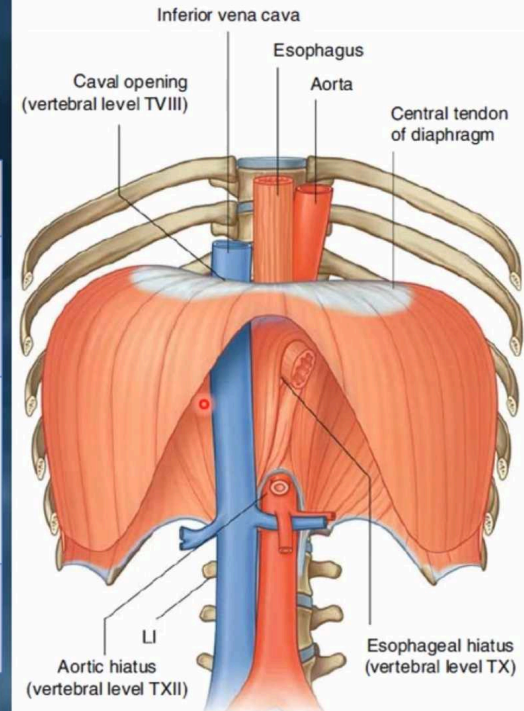
Physiological action of diaphragm

- **During inspiration :**
 - ✓ It **descends** when it contracts, causing an **increase in thoracic volume** by increasing the **vertical diameter** of the thoracic cavity and thus **decreasing intrathoracic pressure**.
- **During expiration:**
 - ✓ It **ascends** when it relaxes, causing a **decrease in thoracic volume** with an **increased thoracic pressure**.



Three Large Openings of Diaphragm

OPENING	LEVEL	STRUCTURES
Vena Caval Foramen	T8	inferior vena cava , and some branches of the right phrenic nerve
Esophageal Hiatus	T10	esophagus , the anterior and posterior vagal trunks , and some small esophageal arteries
Aortic Hiatus	T12	the aorta , the azygos vein , and the thoracic duct



Arterial supply of the Diaphragm

It receives arterial from:

- ✓ The musculophrenic artery.
- ✓ The Pericardiophrenic artery.
- ✓ The superior phrenic artery.
- ✓ The Inferior phrenic artery.

